

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Go, Java, Python, SQL
- **Backend & Systems:** Raft Consensus Algorithm, RESTful APIs, TCP Sockets, FastAPI, Node.js, Express
- **Infrastructure & DB:** AWS (EC2), Docker, PostgreSQL, Git, Linux (WSL)
- **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API, Pandas

TECHNICAL PROJECTS

TitanStore (Distributed Raft Database)

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets

- Developed a fault-tolerant distributed key-value database in Go, implementing the **Raft Consensus Algorithm** for leader election and log replication across a cluster, ensuring high availability.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss, crucial for **real-time anti-fraud systems**.
- Designed and implemented a high-throughput raw TCP server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O, demonstrating **backend services** development.
- Ensured data integrity and reliability through rigorous implementation of consensus protocols, aligning with the need for **validation frameworks** in large-scale data processing.

TitanSwarm (Autonomous AI Co-Pilot)

Jan 2026 - Present

Personal Project · Python, LangChain, FAISS, Streamlit, SQLite, Gemini API

- Architected an autonomous AI Co-Pilot in Python that automates end-to-end job application workflows, enabling the discovery, extraction, and parsing of real-time Software Engineering postings via concurrent JobSpy scrapers across LinkedIn and Indeed.
- Built a zero-hallucination RAG (Retrieval-Augmented Generation) engine using LangChain, Gemini APIs, and a sandboxed FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.
- Designed and implemented a Streamlit Human-in-the-Loop dispatch UI with a Kanban board, enabling users to review, tailor, and manage applications across a full pipeline with persistent state via a SQLite repository layer.
- Applied Test-Driven Development (TDD) with a 82-test suite using Pytest, mocking, and async fixtures to validate all components including the scraper, AI tailor, PDF generator, and repository layer, showcasing strong **testing** practices.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS (EC2), FastAPI

- Migrated full-stack platform from Azure to AWS (EC2), implementing Docker container orchestration to optimize deployments and reduce infrastructure costs, demonstrating experience with **cloud platforms**.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints, contributing to **scalable pipelines**.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025